

the transistor (the round symbol at the centre), the vertical one is the base. Of the remaining two, the one with the arrow mark is the emitter while the other one is the collector. When you are holding the transistor with the flat face upwards and the leads pointing towards you, the leads from the left to right are in the order emitter, base, collector.

To test your creation, take the wires from the output and connect to a headphone or speaker set. If you are using a headphone, one wire should be in contact with the tip of the mini plug while the other is connected to the side. For more stable connections, you may cut off the mini plug and connect to the wires directly.

Switch the circuit on. You should be able to hear a constant pitch hum on the headphone/speaker. Vary the capacitance of the variable capacitance box. Notice that the pitch of the sound changes. This is because the change in capacitance causes a change in the frequency of the generated signal, which in turn causes a change in frequency of the sound, and hence in its pitch.

Alternatively, to test your circuit, you may connect the output to a cathode ray oscilloscope (a device used to form images of waves on a screen). However, we will not explore that option here.